JIMIN LEE

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EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA **Los Angeles, CA**
- Master of Science in Computer Science *Aug 2025 – Expected May 2027*

GACHON UNIVERSITY **Seongnam, Korea**
- Bachelor of Engineering in Software Engineering, *Cum Laude* *Sep 2018 – Feb 2024*

EXPERIENCE

Blue Shield of California **Long Beach, CA**
Service Integration Intern, Integration & Interoperability Platform *Jun 2026 – Aug 2026*
- Built an AI-assisted Boomi-to-MuleSoft migration workflow estimated to reduce medium-complexity migration effort by 70–80%, from ~2 weeks to ~2 days, now adopted for the ~80 remaining integrations
- Developed an LLM agent skill and 6 custom Python MCP tools that trace source-to-target mappings and parse legacy transformation logic, resolving 270+ fields across 15 record layouts
- Migrated 4 enterprise integrations end to end with 4/4Mule packages building and 56/56 MUnit tests passing, auto-generating business-purpose documentation and flagging unconfirmed logic for developer review

University of Southern California, Interaction Lab (Prof. Maja Matarić) **Los Angeles, CA**
Web Team Lead (NIH R01 Funded Project) *Jan 2026 – Present*
- Lead full-stack development of a secure NIH R01-funded longitudinal research platform supporting 90+ participants using Next.js, TypeScript, AWS Amplify, DynamoDB, and Lambda
- Design backend schemas, API contracts, and serverless workflows for longitudinal study data, with Cognito authentication and IAM-controlled service access
- Build LLM-assisted therapeutic reflection workflows and real-time Blossom robot communication through AWS IoT Core and MQTT

University of Southern California, AI for Health Lab (Prof. Ruishan Liu) **Los Angeles, CA**
Graduate Research Intern *Sep 2025 – Aug 2026*
- Developed a cold-start computerized adaptive testing pipeline using LLM-based simulation and IRT calibration; co-authored a paper on synthetically calibrated item banks (under review, ACL Rolling Review)

EMRO – Korea's #1 AI-powered supply-chain software provider **Seoul, Korea**
Full-Stack Engineer, AI Business Division *Sep 2023 – June 2025*
- Developed and launched a production procurement analytics platform for Hyundai Motor Group, translating procurement workflows into quotation search, price benchmarking, and analytics features
- Built enterprise backend APIs using Java/Spring and MySQL/PostgreSQL for validation, analytics, and monitoring
- Integrated a Python/Flask FastText similarity service processing 100K+ quotations monthly and reduced page load time from 10 seconds to under 2 seconds
- Standardized CI/CD across Azure and on-prem environments using Jenkins and Docker with automated health checks and rollback mechanisms

PROJECTS

Peeko – Real-time AI lecture recovery system (Concept & Team Lead) *Apr 2026*
Awards: 1st Place Overall, Build4SC Hackathon (GRIDS x Viterbi, USC, 47 participants)
- Led a 4-person team to design and ship an end-to-end AI product in 12 hours
- Built LLM pipelines for rolling session summarization, "Catch Me Up" recovery, and Q&A detection with a Node.js/Supabase backend

Visurai – AI-powered visual storytelling assistant for dyslexic, ADHD, and visual learners *Oct 2025*
Awards: 1st Place (Context Engineering), Software Engineering Award
USC Good Vibes Only Buildathon (LA Tech Week by a16z)
- Built a FastAPI backend converting text into visual storyboards & narration; streamed results via SSE
- Designed an asynchronous multi-stage generation pipeline with concurrency controls for long-running AI workloads

SKILLS

Languages: Python, Java, TypeScript, JavaScript
Frameworks: Spring Boot, FastAPI, Flask, React, Next.js
Machine Learning & AI: LLM applications, MCP, LangGraph, agentic workflow design, LLM evaluation
Backend & Data: REST APIs, MySQL, PostgreSQL, MuleSoft Anypoint Platform, DataWeave
Cloud & DevOps: AWS, Azure, Docker, Jenkins, CI/CD